

CS100709 - Presentation: A Glimpse of Computer Science

I. Introduction of A Branch in Computer Science

Computer science is a discipline that encompasses both theory and practice, demanding abstract and concrete thinking. The practical aspects of computing are ubiquitous, with nearly everyone being computer users today, and many even functioning as computer programmers. Achieving the desired outcomes from computers necessitates hands-on experience. On a broader scale, computer science is a science of problem-solving. Computer scientists must excel in modeling and analyzing problems while also being proficient in designing and verifying solutions. Problem-solving entails precision, creativity, and meticulous reasoning.

Furthermore, computer science is closely linked to various other fields. Numerous challenges in science, engineering, healthcare, business, and other domains can be efficiently addressed with computers, but it demands computer science expertise and an understanding of the specific application domain. Consequently, computer scientists often acquire proficiency in other subjects.

Lastly, computer science boasts a wide array of specialties, encompassing computer architecture, software systems, graphics, artificial intelligence, computational science, and software engineering. Each specialty area focuses on distinct challenges while drawing from a shared core of computer science knowledge.

Requirements: Work in groups, select one of the computer science topics for a concise introduction (less than 15 slides). The topic can be either specific or broad, but ensure your presentation covers its motivation, research contents, applications, and more. The following are some generic topics for reference:

Big Models 大模型

Hacking and Security 黑客与安全

VR and AR 虚拟现实和增强现实

Artificial Intelligence 人工智能

Computer Architecture 计算机系统结构

Cloud Computing, Internet of Things, and Big Data 云计算、物联网及大数据

Software Engineering 软件工程

Embedded Systems 嵌入式系统

Multimedia Technology 多媒体技术

Computer Networks 计算机网络

Computer Vision 计算机视觉

Open Source Movement 开源运动

Brain-Computer Interface 脑机接口

Techniques: Employ Microsoft PowerPoint, Keynote, or other available software to deliver your presentation.

Evaluation: You will present your topics during the Project Evaluation (refer to your class schedule). Each group's evaluation should not exceed 10 minutes, ensuring everyone has an opportunity to present.

Scoring:

优 汇报组织优秀，逻辑条理连贯明晰，辅有实验或者示例展示，表达生动
良 汇报内容丰富，逻辑清晰，有自主思考和见解，汇报表达准确
中 每人参与完成汇报，汇报内容较为完整，报告形式正确

Bonus: Best presentation materials will be made online.

Notice: Plagiarism is strictly forbidden!